ADLER KIELY

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U.S. Citizen | GPA 4.0 / 4.0 | 40 hrs logged flight time

EDUCATION

Kent State University Kent, Ohio
B.S. Aerospace Engineering *Expected May 2029*
Relevant Coursework: Statics, Calculus I-III, Linear Algebra, Physics I-II (E&M), Engineering Computing (MATLAB)
Honors: President's List • President's Achievement Scholarship • Westinghouse Science Honors Institute

WORK EXPERIENCE

Equinox Engineering Ltd. Canonsburg, Pennsylvania
Engineering Undergraduate Student *June 2026 - Present*
- Reviewed finalized 2D/3D engineering drawings to study company modeling methodology, drafting standards, and geometric dimensioning and tolerancing (GD&T) conventions.
- Gain exposure to engineering, procurement, and construction management (EPCM) workflows across facility design and cross-disciplinary coordination on oil and gas and energy projects.
- Taught myself Microsoft Power BI and built dashboards visualizing engineering project metrics to support data-driven reporting.

Kent State University Kent, Ohio
Academic Tutor - Calculus I & II, Physics I *April 2026 - Present*
- Selected based on 4.0 GPA to tutor calculus and calculus-based physics, breaking analytical problems into approachable steps for peers.

ENGINEERING EXPERIENCE

Kent State University Rocket Team Kent, Ohio
Payload Subsection Lead *August 2026 - Present*
- Direct members of a 6-member payload subsection through design, fabrication, and integration for the 2027 L3 rocket, executing direction from the senior payload lead.
- Developing a 4U CubeSat form-factor weather station payload that measures atmospheric data using pitot tubes and other sensors, with a servo-actuated pitot deployment mechanism.

Aerostructure Member *January 2026 - August 2026*
- Fabricated G10 fiberglass composite fins for a 13-ft high-power rocket, beveling leading edges to reduce pressure drag and epoxy-bonding fins to the body tube for launch and recovery loads.

High-Speed UAV Independent
Sole Designer - In Development *April 2026 - Present*
- Designing a high-speed airframe in SolidWorks and Fusion 360, sizing an integrated electric propulsion package across motor, ESC, propeller, and battery.
- Setting up CFD in ANSYS Fluent on the airframe geometry to estimate drag before committing to a build configuration.
- Prototyping 3D-printed fit-check parts and correcting the model from caliper measurements across successive revisions.

SKILLS & AFFILIATIONS

Design & Analysis: Programming [MATLAB (Proficient), Java (Proficient), Python (Working Knowledge)] • CFD [ANSYS] (Working Knowledge) • CAD [SolidWorks, Fusion 360, Onshape] • Power BI • Excel
Fabrication & Test: Mechanical assembly • Arduino
Flight Systems: Airfoil selection • Propulsion and battery sizing
Aviation: Single-engine land • Airspace classes, sectional charts, weather products, preflight planning
Affiliations: AIAA, Kent State Student Branch (Advocacy Officer, 2026-Present) • Tripoli Rocketry Association (Member) • Lockheed Martin Ethics Case Competition Team, Kent State (Member, 2026)